

Yang Liu

PhD, CEng

Senior Engineer — Digital Tools & Instrumentation (R&D) · Lincoln, United Kingdom

PROFILE

Chartered Engineer (CEng), registered through the Institution of Mechanical Engineers (IMechE), and an industrial R&D specialist working across gas-turbine thermodynamics, performance modelling, data-informed diagnostics, advanced instrumentation and computer vision. International experience in the United Kingdom, Italy and China, connecting physical modelling and experimentation with practical, auditable engineering software.

EXPERIENCE

Senior Engineer — Digital Tools & Instrumentation (R&D)

Jun 2024—Present

Siemens Energy, Industrial Turbomachinery · Lincoln, UK

- Develop digital methods and toolchains spanning thermodynamic performance analysis, data-informed diagnostics and measurement automation.
- Apply computer vision and machine learning to improve the repeatability and turnaround of engineering image analysis.
- Contribute to advanced instrumentation concepts integrating optical, acoustic and vibration measurement for gas-turbine R&D.

Postdoctoral Researcher — Mechanical Engineering

May 2022—May 2024

Politecnico di Milano · Milan, Italy

- Developed an end-to-end wireless measurement chain for rotating machinery, covering sensor interfaces, acquisition, synchronisation and data handling.
- Designed multi-sensor workflows for dynamic strain, vibration and acoustic measurements under a nationally funded research programme.
- Integrated hardware and software architectures and validated experimental repeatability for gear measurement campaigns.

Gas-Turbine Performance & Diagnostics
Researcher / Platform Developer

Nov 2016—Sep 2021

Cranfield University · Cranfield, UK

- Developed algorithms and software modules for gas-turbine performance simulation, gas-path diagnostics, prognostics and creep-life assessment.
- Created a test-data acquisition interface for the PYTHIA platform and validated model outputs against experimental data.
- Produced technical documentation and supported research delivery for an industry-sponsored aero-engine diagnostics programme.

Control & Algorithms Engineer / R&D Engineer — Aerodynamics

Oct 2015—Oct 2016

XAG · Guangzhou, China

- Worked on propeller-motor performance matching, UAV aerodynamics, CFD and aeroacoustic analysis.
- Developed airframe validation rigs and a vibration-damping concept for flight-control instrumentation.

EDUCATION

PhD in Aerospace

Cranfield University · Gas Turbine Performance Group
Thesis: Thermodynamic Cycle Analysis
and Performance Simulation of Reheat Gas
Turbine Engines

2016—2021

MSc Advanced Aeronautical Engineering

Imperial College London · Fluid mechanics and
reduced-order modelling

2014—2015

BEng Aerospace Engineering

University of Liverpool · First Class Honours

2011—2014

Chartered Engineer (CEng)

Engineering Council registration through the
Institution of Mechanical Engineers (IMechE)

CEng

SELECTED
PUBLICATIONS

Off-design performance control and simulation for gas turbine engines with sequential combustors

Yi-Guang Li & Yang Liu · Journal of Power and Energy, 2025 · DOI: 10.1177/09576509251327695

Performance Simulation and Off-Design Performance Control of Reheat Gas Turbine Engines

Yi-Guang Li & Yang Liu · ASME Turbo Expo 2024 · GT2024-124937 · DOI: 10.1115/GT2024-124937

An Integrated Principal Component Analysis, Artificial Neural Network and Gas Path Analysis Approach for Multi-Component Fault Diagnostics of Gas Turbine Engines

O. Alozie, Y.-G. Li, P. Pilidis, Y. Liu et al. · ASME Turbo Expo 2020 · GT2020-15740 · DOI: 10.1115/GT2020-15740

PATENTS

Sensor system for a rotating gear wheel

Published PCT patent application · Inventors: Roberto Palazzetti & Yang Liu · Applicant: Politecnico di Milano · WO2025108683A1 · Public record

Vibration damping component for a drone and drone with the same

Granted Chinese utility model · Inventor: Yang Liu · Applicant: Guangzhou Xaircraft Technology Co., Ltd. · CN205872433U · 2017 · Public record

CAPABILITIES

Gas-turbine engineering

Thermodynamic cycle analysis, performance
modelling, gas-path diagnostics, prognostics,
test validation

Programming & data

Python, MATLAB, C#, C/C++, SQL, data pipelines,
machine learning

Imaging & 3D

OpenCV, Open3D, calibrated imaging, colour science,
visual anomaly detection

Instrumentation

LabVIEW, FlexLogger, Arduino, ESP32, wireless DAQ,
vibration and optical measurement

Simulation

NPSS, Ansys Fluent, STAR-CCM+, Xflow,
XFoil/XFLR5, QPROP

CAD & product development

Siemens NX, Creo/ProE, CATIA V5, SolidWorks,
experimental rig development

ADDITIONAL

Languages

Mandarin — Native or bilingual
English — Full professional
Italian — Elementary

Research themes

Gas-turbine diagnostics, industrial
imaging, advanced instrumentation,
research translation

Outside engineering

RC aircraft modelling, tennis
and cycling